

GENERAL SPECIFICATION G1.5

Painting and Protective Coatings

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G1.5 PAINTING AND PROTECTIVE COATINGS

G1.5.1 Scope

- (a) This Section of the Specification covers surface preparation, painting and protective coating requirements for civil and building works, for iron and steelwork, for pipework (excluding cement mortar lining and factory applied bitumen enamel wrapping which are covered in Pipework general specifications) and for mechanical, electrical and instrumentation equipment and associated steelwork supplied under this Contract.
- (b) Actual locations of the surfaces requiring protective coatings are specified in the particular specification or on the drawings.
- (c) Where not otherwise required by the Specification, or by coating manufacturers' recommendations approved by the S.O., the preparation, painting and protection of surfaces shall be carried out in accordance with the recommendations of BS EN ISO 12944-5, BS EN ISO 12944 Parts 1 to 8, BS EN ISO 14713-1, BS 6150 or BS ISO 27831 PARTS 1 AND 2.

G1.5.2 Reference Standards

- (a) Unless otherwise specified and approved by the S.O., painting and protective coatings, including surface preparation, shall comply with the relevant Singapore Standards and the Reference Standards listed below where applicable:

Standard	Subject
BS 245	Specification for mineral solvents (white spirit and related hydrocarbon solvents) for paints and other purposes
BS EN ISO 1461	Hot dip galvanized coatings on fabricated iron and steel articles. Specifications and test methods
BS EN 12540	Corrosion protection of metals. Electrodeposited coatings of nickel, nickel plus chromium, copper plus nickel and copper plus nickel plus chromium
BS 1336	Knotting
BS EN ISO 4618, BS EN ISO 4618	Paints and Vanishes, Terms and Definitions
BS 7079, BS EN ISO 11124-3	Chilled iron shot and grit for blast cleaning General introduction to standards for preparation of steel substrates before application of paints and related products
BS EN ISO 17834	Sprayed metal coatings
BS EN 10143	Continuously hot-dip zinc coated and iron-zinc alloy coated steel strip, sheet and plate
BS 3416	Bitumen-based coatings for cold application, suitable for use in contact with potable water
BS 4129	Welding primers and weld-through sealants, adhesives and waxes for resistance welding of sheet steel
BS EN 10300	Bitumen-based hot-applied coating materials for protecting iron and steel

Standard	Subject
BS 4652	Metallic zinc rich priming paint (organic media)
BS 7956	Specification for primers for woodwork.
BS 4764	Powder cement paints
BS 4800	Paint colours for building purposes
ASTM B695-04	Standard Specification for Coatings of Zinc Mechanically Deposited on Iron and Steel
ASTM B633-15	Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel
BS 5252	Colour co-ordination for building purposes
BS 7956	Solvent-borne priming paints for woodwork
BS EN ISO 12944 Part 1 to 8	Code of Practice for protective coating of iron and steel structures against corrosion
BS EN 1436, BS EN 1824, BS EN 1871	Road marking materials
BS 6150	Code of Practice for painting of buildings
BS EN ISO 150	Raw, refined and boiled linseed oils for paints and varnishes
BS 6949	Bitumen-based coatings for cold application, excluding use in contact with potable water
BS 7079	Preparation of steel substrate before application of paints and related products
BS ISO 27831 Parts 1 and 2	Code of Practice for cleaning and preparation of metal surfaces
SS 7	Paint - Finishing, gloss enamel
SS 34	Undercoat for gloss enamel
SS 624	Road marking paint

G1.5.3 BIM Submissions by the Contractor

The Contractor is required to incorporate the following information regarding painting and protective:

- (a) Schedule of Protective Coating Systems
 - (i) The Contractor shall prepare a detailed schedule for each element of the Works which shall contain, as a minimum, the following information:
 - 1) System reference;
 - 2) Type of surface;
 - 3) Component or section of the Works;
 - 4) Environment of item;
 - 5) Surface preparation;
 - 6) Paint system or protective coating to be applied;

- 7) Name of coating manufacturer;
 - 8) Brand name and reference number for each coat to be used;
 - 9) Colour of finishing coat stating the Munsell system colour notation;
 - 10) Proposals concerning place of application and detailed method of application of each coat, including information on equipment to be used;
 - 11) Wet film minimum and target thickness for each coat;
 - 12) Dry film minimum and target thickness for each coat; and
 - 13) Density of paint for each coat and coverage of paint per unit volume.
- (ii) The schedule shall be accompanied by manufacturers' data sheets for each coat, including technical description of paint, and the British Standard - BS 4800 shade.
- (iii) The work detailed on the schedule shall meet the requirements specified or otherwise required by the Contract. Where practicable, the schedule shall be in the format of the specimen schedule of paint systems and protective coatings set out in Appendices at the end of this Section. The schedule shall be subject to the approval of the S.O..
- (iv) Protective coatings for surfaces of tanks and other plant or equipment in contact with or subject to corrosive conditions by proximity to chemicals or otherwise in conditions not foreseen in this section of the specification or in BS EN ISO 12944 Part 1 to 8 shall be suitable for those conditions and shall be the subject of design submissions supported by evidence proving satisfactory performance of the proposed coating system(s) elsewhere.
- (b) Record Data for Iron and Steelwork
- (i) For each item or member of iron and steelwork, the Contractor shall provide specific digital record on which the progress of each operation will be recorded from the initial state of the steel before cleaning through to final inspection.
- (ii) All relevant data shall be recorded including:
- 1) Reference to identify the iron or steelwork item or member precisely;
 - 2) Date;
 - 3) Weather conditions (or ambient conditions in a workshop);
 - 4) Preparation, coating or repair being undertaken;
 - 5) Area coated and quantity and type of materials used;
 - 6) Details of tests and measurements taken; and
 - 7) Delays and causes, faults observed and corrections made.
- (iii) The data shall be presented for S.O.'s review and approval not more than 24 hours after the work is completed.

G1.5.4 Materials

(a) Materials - General

- (i) All coatings applied to any part of the Works in contact with water to be used for drinking, washing or cooking shall be non-toxic, non-carcinogenic, shall not impart taste, odour, colour or turbidity to the water or foster microbial growth.
- (ii) Surface protective coatings containing lead shall not be used under any circumstances.
- (iii) Surface protective coatings shall be suitable for use in tropical climates.
- (iv) Surface protective coatings shall be from the product range of reputable suppliers' subject to the approval of the S.O.. All coatings to be applied to a particular surface as part of a specified coating system shall be obtained from a single manufacturer.
- (v) All protective coating systems for concrete, cement mortar rendered and plastered surfaces, woodwork, brickwork and building steelwork shall have a minimum 5 years life to first maintenance.
- (vi) All protective coating systems for structural iron and structural steelwork shall have a minimum 10 years life to first maintenance.
- (vii) All protective external coating systems for iron and steel pipework shall have a minimum 10 years life to first maintenance.
- (viii) As a precedent requirement of approval of the proposed coating system(s) and coating sub-contractor, the Contractor shall deliver to the Board a written guarantee covering the completed coating system in a form satisfactory to the Board. The guarantee shall include defects resulting from materials and workmanship and shall be for a period of five years from the date of completion of Contract.
- (ix) Where priming of surfaces is executed before delivery to Site, the Contractor shall ensure and obtain written confirmation from the appropriate supplier that the primers used in these applications are obtained from the same manufacturer as that approved by the S.O. for the painting or protective coating system.
- (x) Where appropriate the Contractor shall certify to the S.O. that paints, protective coatings or surface treatments supplied for use on the Works are compatible with any timber preservation or fire proofing treatment which may be specified.
- (xi) All surface coating shall be supplied in sealed containers bearing the following information in addition to any statutory requirements:
 - 1) Manufacturer's name, initials or trade mark;
 - 2) Type of surface coating;
 - 3) Whether priming, undercoat or finishing coat;
 - 4) Whether for interior or exterior use;
 - 5) The colour and its reference number;

- 6) The method of application (i.e. brush, spray or roller);
- 7) The batch number and date of manufacture or re-test;
- 8) Storage instructions and shelf life; and
- 9) Appropriate safety instructions.
- (xii) Containers for protective materials other than surface coatings shall bear as much of the above information as appropriate.
- (xiii) Paint containers shall not exceed 20 l capacity except for containers for emulsion, spraying paints, bituminous paints, stone, cement and road marking paints which may be of larger capacity.
- (xiv) All machined, polished or bright surfaces, both external and internal, shall be afforded suitable and adequate protection against corrosion, damage and deterioration.
- (xv) The Contractor shall ensure that, before despatch from the manufacturer's works and after completion of erection, all plant has received suitable preparation followed by the protection system listed in the Appendices at the end of this Section of this specification.
- (xvi) The Contractor shall liaise closely with the plant manufacturer to provide full facilities for the Contractor (or the S.O.) to inspect and check the preparation and surface coating during all stages. The Contractor shall report on his inspections to the S.O..
- (xvii) The Painting Schedules in the Appendices at the end of this Section identify the coating systems to be used and minimum coating thicknesses.
- (b) Protective Coatings for Timber
 - (i) Aluminium Primer: Priming paint for wood shall be aluminium wood primer to BS 7956 except where the S.O.'s approves priming paint to BS 7956. Primer shall be oil modified phenolic resin based incorporating non-leafing aluminium pigments with a solids content by volume of 42%. Minimum Dry Film Thickness (DFT) shall be 30 microns.
 - (ii) Spot Primer: Spot primer shall be inorganic zinc silicate blast primers containing a minimum 60% by weight metallic zinc in the dry film.
 - (iii) Oil Modified Alkyd Resin Undercoat: Undercoat for timber shall be an oil modified alkyd resin single pack high opacity lead free undercoat conforming to SS 34: 1971 with a solids content by volume of 55 to 60%. The flash point shall not be lower than 41°C. Minimum DFT shall be 40 microns.
 - (iv) Oil Modified Alkyd Resin Finish Coat: Finish coat for timber shall be an oil modified alkyd resin single pack enamel conforming to SS 7: 1970, BS 6150 with a solids content by volume of 50 to 55%. The flash point shall not be lower than 41°C. Minimum DFT shall be 30 microns.
 - (v) PVA Copolymer Emulsion Paint: Polyvinyl acetate acrylic copolymer emulsion paint shall be water based single pack non-flammable with a solids content by volume of 45 to 50%. Minimum DFT shall be 50 microns.

- (vi) Thinners and Solvents:
 - 1) The type and quality of thinners to be added to surface coating shall be that recommended by the manufacturer for the particular type of surface coating.
 - 2) White spirit shall comply with BS 245 Type 'A'.
 - 3) Separate supplies of solvents shall be kept on site for brush or other cleaning purposes. Solvents for these purposes shall be tinted a different colour to thinners used for thinning surface coating.
 - 4) Knotting shall comply with BS 1336.
 - 5) Stopping for woodwork, hardboard and plywood shall be a proprietary spirit-based wood filler tinted to match the undercoat colour.
 - 6) Stopping for natural finished woodwork shall be a proprietary spirit-based wood filler tinted to match the woodwork.
 - 7) Linseed oil shall be refined linseed oil and shall comply with BS EN ISO 150.
- (c) Protective Coatings for Concrete, Cement Mortar Rendering and Plaster
 - (i) PVA Copolymer Emulsion Paint
 - 1) Polyvinyl acetate acrylic copolymer emulsion paint shall be water based single pack non-flammable with a solids content by volume of 45 to 50%. Minimum DFT shall be 50 microns.
 - (ii) High Performance Elastomeric Textured Paint
 - 1) Elastomeric textured paint system shall be styrene and acrylic based modified to be waterproof, elastomeric, water vapour permeable and carbon dioxide diffusion resistant.
 - 2) The primer shall be styrene acrylic based modified to be water vapour permeable and carbon dioxide diffusion resistant. Minimum DFT of 25 microns.
 - 3) The undercoat shall be flexible acrylic water based non-plasticised elastomeric modified to be water vapour permeable. Minimum DFT of 350 microns.
 - 4) The finish coat shall be flexible acrylic water based modified to be water vapour permeable, carbon dioxide diffusion resistant, chloride ion diffusion resistant, fungus and algae resistant and ultra violet degradation resistant. Minimum DFT of 50 microns.
 - 5) The water vapour transmission equivalent air layer thickness tested in accordance with ASTM E96-80 shall be less than 4 m for the coating system.
 - 6) The carbon dioxide diffusion equivalent air layer thickness tested in accordance with ASTM E96-80 shall not be less than 50 m for the coating system. Chemical Hardener and Dust Proof Compound.

- 7) Surface hardener for application to hardened concrete or granolithic screed shall be transparent water soluble crystalline material based on magnesium silicofluoride or sodium silicate.
- (iii) Epoxy Coatings
 - 1) Epoxy protective coatings shall be of an approved high build polyamide cured epoxy, 2 or 3 pack, consisting of base and catalyst components completely resistant to the corrosion conditions to be encountered.
 - 2) The primer shall be a low-viscosity, two or three- pack epoxy supplied by the manufacturer of the approved epoxy coating material. It shall have complete compatibility and inter-coat adhesion with the first coat of high build epoxy surface protective coating.
 - 3) The total DFT of the protective coat shall not be less than 750 μm .
 - 4) The epoxy resin coatings shall be so formulated that the epoxy resin content, together with its curing agent, shall be not less than 40% by weight of the solid binder.
- (iv) Glass Flake Reinforced Coatings
 - 1) Glass flake reinforced coating systems shall comprise a two components vinyl ester peroxide hardened catalysed resin prime coat, a silica filled vinyl ester trowel applied base coat, a resin saturated glass fibre reinforcing mat (chopped strand mat) of 0.5 kg/m² weight and a two-component high solid glass flake reinforced vinyl ester resin surface coating as a finish, or equivalent system.
 - 2) The final system coating shall have a thickness not less than 2.5 mm and a surface hardness of 30 Barcol measured in accordance with BS 2782. The detailed system shall be submitted for the approval of the S.O..
 - 3) The glass reinforcement shall be of hardened glass flakes which shall, upon application, self-leaf and laminate to form a barrier of glass within the film. The finish coat shall contain hard alumina fillers in coatings applied to areas of high wear as directed by the S.O..
- (v) Cement Paints
 - 1) Cement paints shall comply with BS 4764 - 1986 - Powder cement paints.
 - 2) Stopping for concrete or sand/cement plastering shall be of similar material to the background and shall have a similar surface finish.
- (d) Protective Coatings for Iron and Steel
 - (i) Protective coatings for iron and steelwork shall comply with BS EN ISO 12944 Parts 1 to 8 for exterior conditions designated as "polluted inland" conditions and the interiors of buildings shall be assumed to be "frequently damp or wet" except for control rooms. The thickness of coatings stated in this section of the specification is the minimum allowable thickness as defined in Clause 19 of BS EN ISO 12944.

(ii) Welding (Blast) Primers for Iron and Steel

- 1) Welding (blast) primers shall comply with BS 4129; they shall be suitable for continuous spray application, dry rapidly, and harden sufficiently within 10 minutes to permit handling of the steel without damage to the blast primer.
- 2) Blast primers shall be non-toxic and shall not give rise to objectionable fumes during flame-cutting and welding. The blast primers, when applied to the steelwork in the thickness of coating necessary to protect the steel effectively during the period before application of the main protective system, shall not affect the strength or quality of welds made through the coating.
- 3) Blast primers for iron and steel shall contain corrosion-inhibiting pigments, adhere firmly to the substrate and form suitable bases for the succeeding coats in the protective paint system.
- 4) Inorganic zinc silicate blast primers shall contain a minimum 60% by weight metallic zinc in the dry film.
- 5) Iron oxide epoxy blast primers shall contain 24% minimum by weight of iron oxide as a percentage of the dry pigment.
- 6) Zinc phosphate/chromate alkyd blast primer shall be fast drying alkyd based blast primer with active corrosion preventing pigments. The drying time at 23°C shall be not more than 15 min to touch dry and 90 min to hard dry in accordance with BS 3900.

(iii) Primers for Iron and Steel

- 1) Primers for iron and steel shall contain corrosion-inhibiting pigments, adhere firmly to the substrate and form suitable bases for the succeeding coats in the protective coating system.
- 2) Primers shall be two pack polyamide cured epoxy zinc phosphate, two pack polyamide cured epoxy zinc chromate, inorganic zinc silicate or micaceous iron oxide epoxy as scheduled.
- 3) Inorganic zinc silicate primers shall contain a minimum 60% by weight metallic zinc in the dry film;
- 4) Micaceous iron oxide epoxy primers shall contain 24% minimum by weight of iron oxide as a percentage of the dry pigment.

(iv) Epoxy Coatings

- 1) The protective coating shall be of an approved two pack high build amine adduct cured epoxy, completely resistant to the corrosion conditions to be encountered.
- 2) The primer shall be a low viscosity high build two pack epoxy supplied by the manufacturer of the approved epoxy coating material. It shall have complete compatibility and inter-coat adhesion with the first coat of high build epoxy surface protective coating.

- 3) Two component high build epoxy coating shall consist of a base and catalyst components. If required approved thinners may be used and shall be applied in strict accordance with the manufacturer's recommendations. Strict control shall be exercised in the issue of thinners for cleaning brushes and equipment.
 - 4) The total DFT of the protective coat shall have a minimum value of 750 microns.
 - 5) Except where otherwise specified epoxy resin coatings shall be so formulated that the epoxy resin content, together with its curing agent, shall be not less than 40% by weight of the solid binder.
- (v) Glass Flake Epoxy Coatings
- 1) Glass flake epoxy coatings shall comprise a two components polyamide cured epoxy resin suitable as a sealer coat, and a two-component high solid glass flake reinforced tar free epoxy surface coating as a finish, or equivalent system. The detailed system shall be submitted for the approval of the S.O..
 - 2) The glass reinforcement shall be of hardened glass flakes which shall, upon application, self-leaf and laminate to form a barrier of glass within the film.
- (vi) Fusion Bonded Epoxy Powder Coatings
- 1) Fusion bonded epoxy coatings described herein are intended for application to ductile iron, cast iron, steel and aluminium substrates used for the fabrication of pipes, fittings and associated components in contact with potable water. Where specified or instructed the ductile /cast iron, steel or aluminium fabricated pipework and other plant shall have a factory applied lining and coating, not less than 300 microns thick, of 100% solids, thermosetting fusion bonded, dry powder epoxy coating.
 - 2) The coating system requirements, performance, testing factory application requirements and practice shall comply with the Water Industry Specification for Polymeric Anti-Corrosion (Barrier) Coatings Ref No FR 0028 with Amendment No 1 dated July 1990.
 - 3) The method of surface preparation shall be as described herein and notwithstanding anything specified elsewhere all surfaces to be fusion bonded epoxy coated shall be blast cleaned to Sa2½ in accordance with BS 7079 Part A1 and immediately coated with a holding blast primer which shall not contain zinc chromate.
- (vii) Chlorinated Rubber Paint
- 1) Chlorinated rubber paint shall be alkyd chlorinated natural rubber based solvent-containing single pack high build with a solids content by volume of 40 to 45%. The flash point shall not exceed 30°C. Minimum DFT shall be 75 microns. The finished coating system shall be vapour permeable.

- (viii) Galvanising
 - 1) Where galvanising is employed it shall comply with the general specifications. All manufacturing processes shall be complete prior to galvanising with all erection markings clearly legible afterwards.
 - 2) Surfaces shall be adequately degreased before the application of any coating, pre-treated with a brush application of an approved etch primer or T wash.
 - 3) Steel subject to hydrogen embrittlement through galvanising shall not be used.
- (ix) Bituminous Coatings
 - 1) Bituminous coatings or bituminous based wrappings shall be formed of the appropriate grades of materials complying with BS 3416 or BS 6949 for cold application, or with BS EN 10300 for hot application.
 - 2) To avoid the possibility of the presence of carcinogenic polyaromatic hydrocarbons all bituminous paints and coatings must be manufactured from petroleum or asphaltic bitumen and not from coal tar bitumen.
 - 3) Where bituminous coatings are to be overpainted, they shall be primed with aluminium sealer.
- (x) Polyurethane Finish
 - 1) Gloss finish coat shall be two pack aliphatic polyurethane to colour as scheduled.
- (e) Miscellaneous
 - (i) Pavement Marking Paints
 - 1) Road, pavement and kerb marking paints shall be chlorinated rubber paints complying with SS 221, thermoplastic material complying with BS EN 1436, BS EN 1824 and BS EN 1871 - Hot-applied thermoplastic road marking materials- applied in compliance with the PWD general specification Clause 17.5.

G1.5.5 Painting of Plant, Equipment and Pipework

- (a) All plant, equipment and pipework shall be painted over the entire surface area, where otherwise specified or agreed with the S.O..
- (b) All finish coat colours shall comply with BS 4800. Finish coats shall be in accordance with the Appendices of Colour Finishes included at the end of this Section or as directed by the S.O.. Each coat shall differ noticeably in tint from the previous coat.
- (c) Unless otherwise specified the colour coding for plant, equipment and pipework shall be as given in the Appendices at the end of this section.
- (d) Pipework including pipelines, valves, instruments and associated equipment shall be identified for area, system type etc. by the use of identifying bands and tags over their entire length in accordance with the general specifications.

- (e) As well as being painted, all pipework and ductwork (whether insulated or not) shall have identification discs or bands generally as BS 1710, Annex D - Code colours for General Building Services and SS 553 Code of practice for air -conditioning and mechanical ventilation in buildings Annex B identification of pipelines (colour code).
- (f) Finish coats shall be applied as specified in the Appendices of Colour Finishes in accordance with the requirements of BS 1710, BS 4800, BS 5252 and BS EN ISO 7010.
- (g) Items which do not required to be painted shall include:
 - (i) Insulated ductwork with decorative aluminium or stainless steel finish;
 - (ii) Copper pipework and fittings; (except where specifically stated);
 - (iii) Galvanized pipework, ductwork, conduit, trunking or cable trays where concealed within duct shafts or false ceilings;
 - (iv) Materials with a factory applied anodized, baked enamel or painted finish, provided that the colours are approved by the S.O. prior to application; and
 - (v) Insulated pipework or ductwork concealed within duct shafts or false ceilings.

G1.5.6 Workmanship - General

- (a) The Contractor shall follow the requirements of the coating system manufacture with regards to:
 - (i) Storage of materials;
 - (ii) Environmental control;
 - (iii) Preparation of surfaces;
 - (iv) Priming or sealing;
 - (v) Number, order and thickness of prime coats and method of application; and
 - (vi) Finish coat and method of application.
- (b) Storage of Materials
 - (i) Surface protective coating materials shall be stored in sealed containers in a lock-up and shaded store maintained at a temperature of not less than 4°C and not more than 35°C. Any special storage conditions for the surface protective coating materials recommended by the manufacturer shall be observed.
 - (ii) A separate store shall be provided for cleaning solvents used for brush or other cleaning purposes. Cleaning solvents shall not be stored where surface protective coating or thinners are stored or where surface protective coating is prepared for application.
 - (iii) All surface protective coating materials shall be supplied from the Contractor's store to the applicators ready for application. Any addition of thinners shall be made in the store under supervision and up to the limit detailed on the manufacturer's data sheet for the particular method and conditions of application concerned. Surface protective coatings shall be thoroughly stirred before and during use, and shall be strained as and when necessary.

- (iv) Containers of any one surface protective coating shall be used in strict order of delivery from the manufacturer.
 - (v) No surface protective coating shall be used more than 12 months after the date of its manufacture unless it is demonstrated by the contractor to the satisfaction of the S.O. that the quality is unaffected. No surface protective coating shall be used after the expiry of the shelf life specified on the surface protective coating containers.
 - (vi) At the end of each working period during which coating application has been carried out, all two-pack primers and similar chemically-cured surface protective coatings with a limited pot life which have been mixed but not used, shall be discarded.
 - (vii) Other types of surface protective coating from painters' kettles shall be returned to the store and kept in sealed containers with not more than 10% usage.
- (c) Environmental Control
- (i) Surface protective coating shall not be applied under adverse conditions. Air temperatures shall be above 5°C and relative humidity as specified by the surface protective coating manufacturer applied strictly in accordance with manufacturers' instructions.
 - (ii) The Contractor shall ensure that, for the materials specified, application conditions are in accordance with the specification and manufacturer's instructions, surface protective coating only being applied to surfaces that have been cleaned and prepared in accordance with these instructions.
 - (iii) Where local climatic conditions are such that the specified requirements are difficult to achieve the Contractor shall provide temporary protection.
 - (iv) Surface protective coating shall be applied only when the relative humidity is below 95%. Surfaces to which the surface protective coating is applied shall be dry and shall remain free from moisture until the coating has dried or cured sufficiently to avoid effects harmful to the subsequent appearance or satisfactory protective behaviour of the surface protective coating.
 - (v) Coating operations shall not be carried out in the vicinity of other operations which might cause dust.
 - (vi) The Contractor shall not dispose of solid or liquid wastes from surface protective coating operations down any permanent sanitary fittings, sinks or drains and a non-inflammable portable container shall be provided and used for this purpose.
 - (vii) The Contractor shall take every precaution to protect newly surface protective coated surfaces from damage by any cause including wind-blown dust. Precautions shall include warning signs, barriers and covers.

G1.5.7 Preparation of Surfaces

- (a) Protection of Surfaces
 - (i) All necessary precautions shall be taken by the Contractor before the commencement of any surface protective coating activities to ensure that no damage or disfigurement of any kind is caused thereby to any part of the Works. If in the opinion of the S.O., the precautions taken are inadequate the Contractor shall make all such improvements as the S.O. directs.
 - (ii) The Contractor shall exercise particular care not to apply surface protective coating to any surfaces not intended to be coated and shall promptly and thoroughly remove any surface protective coating applied to such surfaces. All floors and surfaces liable to damage or spillage shall be protected with dust sheets.
 - (iii) No manufacturer's name plate identification, vented filler plugs in gearboxes or grease nipples shall be painted over.
 - (iv) The Contractor shall remove all ironmongery and door furniture before surface protective coating work commences and refix all items on completion of the surface protective coating.
- (b) Safety Precautions for Surface Protective Coating Operations
 - (i) The implementation of manufacturers' recommendations concerning health and safety aspects of all surface protective coatings used on the Works shall be mandatory under this Contract.
- (c) Preparation of Surfaces for Protective Coating
 - (i) The Contractor shall regard the preparation of surfaces to be protective coated as work of fundamental importance, the object being to ensure the production of sound, clean, smooth and dry surfaces. The preparation shall have no detrimental effect on the substrate to be coated nor shall it be prejudicial to subsequent coating operations.
- (d) Preparation of Wood Surfaces
 - (i) The preparation of wood surfaces shall include stopping, filling, rubbing down and bringing to a smooth surface free of dust.
 - (ii) Surfaces which are to be surface protective coated shall be rubbed down to remove all contaminating substances and imperfections which would be visible in the finished surface protective coating film. The surfaces of knots and resinous streaks shall then be painted with two coats of shellac knotting to BS 1336, the first being allowed to dry before the second is applied.
 - (iii) All protruding nails, screws and metallic inserts shall be driven down below the surface and the hole stopped with putty.
 - (iv) The surfaces of timber treated with a water-borne preservative by an impregnation process shall be rubbed down and dry-brushed to remove all traces of efflorescence before the primer is applied.
 - (v) Where surfaces are suspected of being infected with mould, they shall be thoroughly treated with a fungicide.

- (vi) Wood surfaces shall not be surface protective coated when the moisture content of the timber measured with an electric conductivity moisture meter exceeds 12 percent for interior surfaces or 17 percent for exterior surfaces.
- (vii) The backs of timber members in contact with walling shall be surface protective coated with two coats of aluminium wood priming surface protective coating prior to fixing.
- (e) Preparation of Plaster, Brickwork and Concrete Surfaces
 - (i) Efflorescence present on the surfaces of plaster, brickwork and concrete shall be removed by scraping, brushing and wiping with a dry coarse cloth followed by a damp cloth. When efflorescence has been removed surfaces shall be left for at least three days and approved by the S.O. before surface protective coating. Surface protective coating shall be deferred where further salt deposits form on the surface during this period.
 - (ii) Plaster surfaces to be surface protective coated shall be cleaned down, filled and smoothed as necessary.
 - (iii) Brickwork, blockwork and concrete surfaces shall be cleaned of all contaminating matter before being surface protective coated. Subject to the approval of the S.O. large holes which would cause a break in the surface protective coating film shall be filled with mortar, with the surface being rubbed down to match the surrounding areas.
- (f) Preparation of Concrete and Rendered Surfaces
 - (i) Concrete and rendered surfaces to receive protective coating shall be thoroughly cured before the application of an epoxy filler primer or surface protective coating is begun.
 - (ii) All surfaces to be protected with a protective or decorative coating system shall be prepared as described below, maintaining the sequence of operations indicated.
 - (iii) Areas of concrete contaminated with machine oil or grease shall be cut out as necessary to remove all traces of such substances and shall be made good with an approved epoxy mortar.
 - (iv) Areas contaminated with release agent shall be scrubbed with suitable emulsion cleaners and any mould growth shall be treated with water-soluble fungicide. All surfaces so treated shall be rinsed off with water until they are clean of any mould growth and any other water soluble on the surfaces (e.g. salt).
 - (v) All concrete and rendered surfaces to be coated shall be lightly blast cleaned to remove the cement-rich surface layer, taking care not to roughen the surface unduly. Grit and detritus shall be removed by vacuum cleaning immediately prior to priming.
 - (vi) Blow-holes and honeycombed areas in the concrete which in the opinion of the S.O. are not capable of being satisfactorily levelled at the primer stage shall be filled with an epoxy mortar supplied by the coating manufacturer for the purpose. Such mortar shall be knifed into the surface to level the area and to leave no excess.

- (vii) Concrete surfaces which are to receive glass fibre/glass flake reinforced vinyl ester coatings shall be abraded by sand blasting and/or scarifying and shall have been allowed to cure for at least 28 days.
- (g) Preparation of Miscellaneous Metalwork, Fixtures and Fittings for Building Works
 - (i) The primer coat shall be applied to steelwork as soon as practicable after fabrication and, where possible, shall be of the same general formulation as that used for the prefabrication primer, if one has been used.
 - (ii) The preparation of ferrous metal surfaces primed at works shall include the removal of any rust and touching up with primer within seven days of delivery to Site.
 - (iii) The preparation of ferrous metal surfaces delivered to Site unprimed shall include removing all mill scale and thoroughly wire brushing before surface protective coating.
 - (iv) Steel which is to be totally embedded in concrete shall not be primed.
- (h) Preparation of Iron and Steel by Blast Cleaning
 - (i) Surfaces shall be blast cleaned to BS 7079: Part A1, Grade Sa1, Sa2, Sa2½ or Sa3, as specified, or otherwise as required or recommended by the Reference Standards and as proposed by the Contractor and approved by the S.O..
 - (ii) The period and conditions under which steel stocks are held and the speed, conditions and methods of fabrication shall be such that immediately before blast cleaning no significant part shall have rusted more extensively than is shown for rust grade B in BS 7079: Part A1.
 - (iii) Where not otherwise required (e.g. to give a rougher surface to receive sprayed metal coatings), the abrasive used shall be chilled iron grit to BS 7079 E3, (or a steel grit of similar hardness), preferably all passing a No. 30 (0.5 mm aperture) mesh and all retained on a No. 36 (0.42 mm aperture) mesh but a finer grit to BS 7079 E3 Table 2 Grade G07 would be an acceptable alternative. The abrasive shall not be re-used without satisfactory cleaning.
 - (iv) Surfaces to be blast cleaned shall be prepared by thoroughly removing all dirt, oil, grease and other contaminants by one of the methods recommended by BS ISO 27831 Parts 1 and 2, using an approved emulsion cleaner which shall be changed regularly.
 - (v) Blast cleaning shall be carried out only when the ambient temperature is above 5°C and the relative humidity is below 85%. Blast cleaning shall be carried out in conditions under which no protective coatings can fall onto or condense on the surfaces during or immediately after cleaning. Special attention shall be paid to the cleaning of any ends, re-entrant corners and edges which will not be cut off or welded after blasting.
 - (vi) After blast cleaning, dust and blasting grit particles shall be removed from the surfaces, where possible by vacuum cleaning. Any lamination or other defects in the surfaces exposed by blast cleaning, which could form a corrosion centre under subsequent protective treatment cover (but which is not otherwise a cause for rejection of the item), shall be marked clearly for later remedial work.

- (vii) Blast cleaning shall be followed within four hours by the application of a blast primer, the priming coat of the protective coating system or hot dip galvanising.
- (viii) The Contractor shall supply and operate such dehumidification equipment as may be necessary to preserve blast cleaned surfaces in a pristine condition until they can be coated and/or to provide the curing conditions necessary for such coats.
- (ix) Steelwork fabricated under factory conditions shall not be blast cleaned until all machining and fabrication has been completed, with such fabrication comprising continuous welded runs with all exposed surfaces accessible for subsequent treatment. All slag and spatter shall be removed from the area of welds by chipping hammer prior to blast cleaning.
- (x) Where surface defects revealed or present during blast cleaning are subsequently ground out or otherwise made good, the resulting dressed surfaces shall be blast cleaned so that the preparation grade characteristics of the entire surface are uniform.

G1.5.8 Application

- (a) Application - General
 - (i) Application of surface protective coating other than hot dip galvanising shall be by one of the following methods, the choice being the Contractor's unless one particular method is specified or is recommended in the manufacturer's instructions for use of the surface protective coating:
 - 1) Brush;
 - 2) Air pressure spraying;
 - 3) Airless spraying; or
 - 4) Roller.
 - (ii) Coatings with suitable flow-out properties may be applied to large flat areas by mohair sleeved or similar approved type of rollers. Rollers shall not be used under any other circumstances or for the application of paints containing micaceous iron oxide pigments.
 - (iii) All surface protective coatings shall be prepared and applied in accordance with the manufacturer's instructions. Copies of appropriate data sheets and of the relevant parts of this Specification shall be issued to all the supervisors and foremen concerned with surface preparation and coating. Where a manufacturer's instructions conflict with this Specification, a ruling shall be sought from the S.O..
 - (iv) No thinners or cleaners shall be employed other than those recommended by the surface protective coating manufacturer.
 - (v) The coating manufacturers printed instructions concerning hot weather application shall be strictly adhered to.

- (vi) Unless otherwise recommended by the manufacture, at least 24 (twenty-four) hour shall elapse between the application of successive coats. The elapsed time shall not exceed the following:
 - Between application of primer (not blast primer)21 days and next coat
 - Between application of prime coats14 days
 - Between application of finishing coats2 days
- (vii) Each coat shall be touch dry before the next coat is applied and each coat shall be rubbed down with fine glass paper before the next coat is applied.
- (viii) Where this elapsed time is exceeded, the surfaces to be re-coated shall first be suitably abraded or treated with solvent to remove gloss and to give a good key all at the expense of the Contractor.
- (ix) The dried films of the specified surface protective coatings shall be free from bloom, sinkage, sheerness, wrinkling, sagging, curtaining, discolouration and extraneous matter.
- (x) No exterior or exposed surface protective coating work shall be carried out under adverse weather conditions.
- (xi) Surface protective coating shall insofar as is practicable be shaded from direct sunlight to prevent wrinkling and blistering. Whenever possible, exterior surface protective coating operations during the day shall be so programmed as to be carried out in shadow.
- (xii) Surface protective coating of surfaces shall not be carried out when the relative humidity of the atmosphere is such that condensation is present on such surfaces or the application and/or drying of the surface protective coatings is likely in the opinion of the S.O. to be affected.
- (xiii) Brushes, tools and equipment shall be kept in a clean condition and surfaces shall be clean and free from dust during surface protective coating. The final coat shall be uniform in colour and free from brush marks, runs or other defects.
- (xiv) The paint DFT shall be measured by a coating thickness gauge.
- (b) Coating of Building Works Surfaces
 - (i) Surface protective coating work shall comply with the recommendations of BS 6150 and the instructions of the surface protective coating manufacturer, except where otherwise approved by the S.O..
 - (ii) All wall and ceiling surfaces scheduled to be surface protective coated with emulsion paint, shall receive at least three coats. On absorbent surfaces the first coat of emulsion may be thinned with the prior approval of the S.O.. At least 4 hours shall be allowed between the applications of successive coats of emulsion.
 - (iii) The Contractor shall satisfy the S.O. as to the covering capacity of paints and emulsion according to the absorption of varying surfaces. The method and application of surface protective coatings and emulsions shall be sufficient to give solid cover in the number of coats specified.

- (iv) The Contractor shall prepare sample panels of surface protective coating on internal walls and ceilings and sample areas of surface protective coating to metal surfaces and timber surfaces for the S.O.'s approval.
- (v) The Contractor shall clean off all paint stains from joinery, metalwork and floors on completion of the surface protective coating work.
- (c) Surface Protective Coating to Iron and Steel
 - (i) Primer
 - 1) The main priming coat shall be applied to the iron or steel as soon as practicable after fabrication and where possible shall be of the same general formulation as that used for the blast primer, if one has been used.
 - 2) After blast priming of any steel plate or section, edges, corners, crevices or holes still present after fabrication, and not part of a welded joint or interior surface of a hermetically sealed void, shall be given an additional stripe coat of the appropriate main primer by brush to ensure continuity of protection of the steel across such edges and the like.
 - 3) Where blast cleaning is impracticable the inside surfaces of members and similar non-fabricated sections shall be cleaned to preparation grade St 2 as shown in BS 7079: Part A1. The priming coat shall be compatible with this preparation grade.
 - (ii) Bolted Connections
 - 1) Joint areas for bolted connections shall be masked to maintain the surfaces free from any surface protective coating applied prior to making the connection. Masking shall be removed before erection.
 - 2) After installation and after all bolts have been tightened, the areas of the connection shall be cleaned to remove all dirt, dust, oil or other contaminant. Particular care shall be taken to ensure that all traces of oil and grease are removed from bolts, nuts and washers.
 - 3) Any surrounding areas of surface protective coating work which have been damaged shall be repaired.
 - 4) Bolts, nuts and washers and any areas exposed at bolted connections shall also be primed as specified, particular care being taken to ensure that any crevices are fully sealed.
 - 5) The remaining coats of the surface protective coating system shall then be applied.
 - (iii) Welds
 - 1) After the completion and grinding of exposed welds, the weld spatter, weld residues and all alkaline deposits and contaminants shall be removed from the surface of the steel-work, and welds and any other areas affected or damaged by the welding process shall be blast cleaned as herein specified.

- 2) Blast primer shall be applied to the blast cleaned areas as specified and the remainder of the surface protective coating system applied as necessary to complete the protection across the welded seam and damaged areas to the same state as that existing generally on the faces of the parent metal.
 - 3) Each coat applied shall overlap the corresponding existing coat by at least 50 mm both sides of the weld on each exposed face of the parent metal.
 - 4) Internal spaces which cannot easily be field painted, shall be painted prior to closing them up in the shop. Surface preparation, shop painting and field painting shall be as specified herein.
- (iv) Repairing Damaged Coats
- 1) Factory prepared surfaces, other than zinc coated surfaces, damaged during site fabrication and/or handling shall be treated as described in the following paragraphs.
 - 2) All slag and spatter shall be removed from areas of welds by chipping hammer and such areas together with all other areas of damaged priming or other coat shall be thoroughly mechanically wire brushed and given one coat. The priming coat shall be applied by brush, taking care to cover uneven surfaces completely, particularly those of welds. Further approved coats shall be applied where necessary to build up the thickness of the repaired coating to at least that of the original coating.
 - 3) Areas of surface protective coating on steelwork which have been damaged shall be cleaned to sound material and the edges of the undamaged surface protective coating shall be smoothed with sandpaper to a gentle bevel. The specified surface protective coating system shall then be applied to bring the damaged area up to the same state of protection as the surrounding surface protective coating work, with each coat of new surface protective coating overlapping the corresponding existing coat of surface protective coating by at least 50 mm.
 - 4) Where the steel is exposed, it shall be cleaned to remove all corrosion, by blast cleaning if ordered by the S.O..
 - 5) Where epoxy coatings are damaged, suitable repair material supplied by the manufacturer of the original coating shall be applied.
 - 6) Damaged areas of zinc based coatings not individually exceeding 40 mm² in area shall be made good by the application of a metallic zinc-rich primer complying with BS 4652. Sufficient material shall be applied to provide a zinc coating at least equal in thickness to the original coating.

(d) Epoxy Coating Work

(i) General

- 1) In addition to the general requirements of this specification the following conditions shall apply wherever epoxy coating work is being carried out:
 - The work shall be illuminated to the satisfaction of the S.O.;
 - Forced draught ventilation to the approval of the S.O. shall be used wherever required for the needs of personnel or for drying out surfaces; and
 - Operatives shall work in pairs.
- 2) The Contractor shall demonstrate all his methods, equipment and materials before any work commences. Sample areas of substrate shall be prepared and coated as required by the S.O. and for his approval.
- 3) There shall be strict control of surface cleanliness between primer and coating and between coats of the same. Vacuum removal of dust and sand shall be employed and contamination shall be removed as specified in appropriate preparation clauses herein.
- 4) Where dirt or dust has become trapped in the epoxy coating surface it shall be removed with suitable abrasive paper. The surface being coated shall be free of visible moisture throughout these operations.
- 5) The primer shall be applied by suitable nylon bristle brush or spray over the whole area to be coated at such thickness that it may then be squeegeed into the pores of the concrete.
- 6) Any excess shall be effectively removed before application of the high build epoxy.
- 7) The S.O. may approve an alternative application method where the Contractor can demonstrate a suitable technique.
- 8) Epoxy coating shall be applied only to clean dry primed or previously coated surfaces. Any thick runs or collections of surface protective coating shall be removed before they harden.
- 9) No fewer than two coats shall be applied over the primer by airless spray and no fewer than three by brush.
- 10) Each coat shall be distinctly different in colour from the primer or previous coat. Each coat shall be seen to have completely covered the preceding coat without 'misses' or pinholes or any areas visibly low in thickness. A high voltage pinhole detector shall also be used to determine the integrity of the coats.
- 11) The manufacturer of the coating shall stipulate primer and epoxy re-coating intervals for all curing temperatures likely to be encountered and these shall be adopted with a maximum tolerance of +4 hours. Where this is exceeded, the surfaces to be re-coated shall first be suitably abraded or treated with solvent to remove gloss and to give a good key.

- 12) Wet thickness gauges shall be used continually by the coating operators to check that sufficient coating is being applied to achieve the desired DFT.
 - 13) Containers of epoxy coatings shall be of a pack-size suitable for complete usage by one operator within the pot life of the material at the highest likely ambient temperature.
 - 14) The S.O. may approve the supply and use of larger packs but only where the proportioning of the various parts, their mixing, and issue of coatings shall be under the direct control of a skilled and experienced operator.
 - 15) A mechanically operated mixer shall be used and shall be of a design which does not entrain air in the coating.
 - 16) The pot-life of epoxy primers and coatings when mixed shall be not less than one hour at the temperature at the place of application at the time of mixing. When approving paints, the S.O. will have regard to the ambient temperatures likely to obtain for field application and may restrict coating application to those times when the temperature is sufficiently low for the specified pot-life to be ensured.
- (ii) Application of Fusion Bonded Epoxy Coatings
- 1) Welds shall be fettled smooth, edges and corners shall be rounded to 1.5 mm minimum radius, slivers and scabs shall be removed.
 - 2) The surface shall be tested for soluble salts and reblasted if soluble salts are found to be present. All traces of grit, dust, oil, grease and other contaminants shall be removed and coating commenced before formation of visible oxidation of the surface.
 - 3) The metal shall be pre-heated to a temperature recommended by the manufacturer, the epoxy powder applied by immersion in a fluidised bed following which excess powder shall be removed. The powder shall be allowed to flow out completely before curing.
 - 4) The minimum thickness shall be 300 microns. The maximum thickness shall be 1.6 mm on the pipe barrel and 2.0 mm on flanges and appurtenances.
 - 5) The thickness of the coating, including any repaired areas, shall be checked with a calibrated tester.
 - 6) Spark testing, for pinholes, voids, contamination, cracks and damaged areas, shall be carried out using a high voltage spark generator. All coatings shall be examined to a Class A coating classification. Any fitting with 2 or more holidays per 3 square metres of surface or any pipe with 30 or more holidays per pipe length shall be rejected and the surface stripped, blasted and recoated.
 - 7) Repairs due to coating imperfections or damage shall be undertaken using a brush applied compatible two-pack liquid epoxy compound.

- 8) The area to be repaired shall be cleaned to remove dirt, grease, scale and damage coating, which shall be feathered. Pinhole surface preparation is not required other than removal of detrimental contaminants which impair the adhesion of the repair material.
 - 9) The surface coating repair shall be applied by means of an approved applicator.
- (iii) Application of Epoxy Coatings to Concrete and Rendered Surfaces
- 1) No priming shall commence until the moisture content of the cementitious surface is less than 5% measured by "Wet Check" moisture meter or other instrument approved by the S.O..
 - 2) Similarly, moisture measurements over the primer or any epoxy intercoat shall not exceed 1% on the concrete scale of the instrument when the probe tips are held against such coated surfaces just prior to re-coating.
 - 3) Wherever the surface protective coating inspection gauge has been used, and wherever the coating has been otherwise damaged, the surface shall be abraded for 50mm around such damage and the area shall be touched in with not less than two thick applications to restore the coating integrity and thickness.
 - 4) Adhesion tests shall be carried out on the cured coating surface using the test equipment supplied under the Contract. The resulting test specimens shall show no indication of poor adhesion to the substrate, residual laitance or intercoat adhesion weakness.
- (iv) Application of Glass Fibre/Glass Flake Reinforced Vinyl ester Coatings
- 1) The prepared surface shall be coated with one coat of a sealing primer followed by one coat of vinyl ester/peroxide primer at a DFT of 50 microns and allow to cure tack free. The base coat shall be trowel applied to a thickness of 1.5 mm followed immediately by application of the chopped strand mat glass fibre reinforcement.
 - 2) The mat is to be saturated with catalysed resin and rolled with a ribbed roller to remove all trapped air. After the resin has hardened the glass flake vinyl ester top coat is to be spray applied to a DFT of 450 microns.
 - 3) The manufacturers' instructions regarding storage, mixing, application and ventilation must be strictly followed. No source of ignition must be allowed within 20 metres of the working area until the coating system has cured.
 - 4) Adhesion tests shall be carried out on the cured coating surface using the test equipment supplied under the Contract. The resulting test specimens shall show no indication of poor adhesion to the substrate, residual laitance or intercoat adhesion weakness.
 - 5) At any interface of glass flake polyester surface protective coating system with an epoxy coating system the glass flake polyester shall be applied first with a 100 mm overlap.

(e) Inspection

The following tests shall be carried out:

(i) Checking Wet Film Thickness

- 1) Wet film thickness gauges of an approved type shall be provided to check the rate of surface protective coating application.

(ii) Checking DFT

- 1) The thickness of the built-up dry film after each surface protective coating coat applied to steel or other magnetic surfaces shall be measured systematically, with a DFT gauge.

(iii) Holiday Detection on Steel and Iron Surfaces

- 1) The Contractor shall use a suitable method of detecting pinholes in the coating system after trials on test plates. The sweep voltage on high voltage dc equipment shall not exceed half the voltage required to spark through the complete surface protective coating system specified.

(iv) Adhesion of Reinforced Vinyl Ester Resins to Concrete

- 1) The Contractor shall conduct adhesion tests on the cured reinforced vinyl ester coating to demonstrate an adhesion not less than 1.5 N/mm² and a hardness of 30 Barcol.

(f) Defects

- (i) Defects are defined in BS EN ISO 4618 "Glossary of Painting Terms".

- (ii) The Contractor shall ensure that all coatings are free from defects and adequate in all respects for the purpose intended.

- (iii) The surface protective coating system shall be deemed to have failed if:

- 1) After surface protective coating, damage has been caused by handling, impact, abrasion or welding;
- 2) Any portion of the surface protective coating film separates from any other or the parent metal; and
- 3) After surface protective coating the total DFT measured by the coating thickness gauge is less than that specified.

- (iv) Failure shall not include:

- 1) Loss of gloss; and
- 2) Variation of shade, not affecting the anti-corrosive properties of the system.

G1.5.9 Special Applications

- (a) Galvanising and Other Metal Coatings
 - (i) Where steel or wrought iron is to be galvanised, the galvanising shall be executed after all fabrication has been completed. Surfaces shall be prepared by removing all weld slag and spatter, surface protective coating, oil and grease.
 - (ii) The articles shall then be shot/grit blasted to SA2½. Galvanising shall be carried out in accordance with BS EN ISO 1461 so that the whole of the metal is evenly covered, with clean edges and bright surfaces being presented.
 - (iii) The minimum average coating weight shall be not less than 610g/m² of galvanised surface except for tubes to BS EN 10255 and sheet, plate or strip to BS EN 10143, which shall be galvanised as specified in those standards.
 - (iv) On steel of under 5 mm thickness, lower coating weights in conformity with Table 1 of BS EN ISO 1461 will be accepted only where they and any overcoating to be applied will provide the required life to first maintenance, as specified in BS EN ISO 12944 Parts 1 to 8.
 - (v) Zinc or cadmium coatings applied by electroplating or Sherardizing shall be used only on small components where galvanising would be impracticable and with the approval of the S.O..
 - (vi) Zinc based protective coatings on threaded components shall be applied only by Sherardizing which shall comply with BS EN ISO 1456 for service condition No. 4. Sherardized coatings shall be Class 1 to ASTM B695-04 or ASTM B633-15.
 - (vii) The application of zinc based paints as touch up to hot dip galvanised metalwork shall not be carried out.
 - (viii) Cleaning galvanised surfaces prior to coating:
 - 1) Dirt and other adherent contaminating matter shall be removed from galvanised surfaces by brushing with wire or stiff bristle brushes or by other methods approved by the S.O.. Cleaning shall be such as not to damage the galvanising.
 - 2) Oil or grease shall be removed from galvanised surfaces by the application of an emulsion cleaner. Any zinc corrosion products remaining shall be removed by washing with water and scrubbing with hard bristle brushes.
 - 3) After being cleaned and degreased all galvanised surfaces shall be etched with T-wash as described in BS EN ISO 12944 Parts 1 to 8. If any surface fails to turn black, the cleaning, degreasing and etching processes shall be repeated.
- (b) Coating System for Electrical Equipment Enclosures
 - (i) The enclosure shall be given its protective coating at the place of manufacture and before installation of its internal electrical fittings.

- (ii) Electrical panels, including switchboards, control panels and instrument panels, installed within buildings shall be prepared as follows:
 - 1) Thoroughly clean surfaces to remove rust, scale, dirt, loose paint etc. and degrease by the use of solvents which are compatible with the paint finish to be applied. If rust proof steel has not been used in the construction, the surfaces shall be treated with a passivating agent such as phosphoric acid.
 - 2) Internal surfaces shall have a minimum of five coats of paint of which the first shall be an approved priming coat, the second and third a suitable undercoat, all of which shall be rubbed smooth when dry before application of the next coat. The colour of undercoat paints shall be different from priming and finishing coats.
 - 3) External surfaces shall have a minimum of five coats of paint of which the first shall be an approved priming coat, the second and third a suitable undercoat, all of which shall be rubbed smooth when dry before application of the next coat.
 - 4) Final coats shall be of stove enamel paint to a finish and colour as specified in the Appendices. The DFT shall be not less than 100 microns.
 - 5) Electrical equipment enclosures which are to be installed in exposed positions or in damp conditions shall receive a surface preparation containing zinc prior to application of the primer undercoat and finishing coats.
- (iii) A 500 ml tin of matching touch up paint shall be supplied with each panel.
- (c) Protective coatings for Fuel Oil Storage Tanks

The fuel oil storage tanks shall be subject to surface preparation and protective coating application generally as follows:

 - (i) External Surfaces
 - 1) Grit blast to BS 7079 and BS EN ISO 8503 second quality or Swedish Standard 2.5 to a blast profile of 50 - 70 microns;
 - 2) Apply one coat of an inhibited iron oxide pigmented two-pack epoxy blast primer to a DFT of 25 microns;
 - 3) Apply one coat of a two-pack zinc chromate or zinc phosphate primer to a DFT of 75 microns;
 - 4) Apply one coat of a two-pack high build epoxy chemical resistant intermediate system to a DFT of 100 microns; and
 - 5) Apply one coat of a two pack cured polyurethane gloss finish to a DFT of 40 microns.
 - (ii) The colour of the paint shall comply with the requirements as given in the general specifications.
 - (iii) Internal Surfaces
 - 1) Grit blast to BS 7079 and BS EN ISO 8503 second quality or Swedish Standard 2.5 to a blast profile of 50 - 75 microns;

- 2) Apply one coat of a two-pack chemically cured holding primer to a DFT of 25 microns; and
 - 3) Apply one coat of a suitable two pack lining to a DFT of 125 microns.
- (d) Other
 - (i) GRP Laminates and Mouldings
 - 1) GRP covers and guards shall be pigmented to give the finished colour without painting.
 - (ii) Air Conditioning Pipes
 - 1) All black steel pipes shall be coated with two coats of zinc chromate primer followed by one coat of bitumastic paint. The coated pipe shall be wrapped in vapour barrier insulation set with a suitable adhesive.
 - 2) The vapour barrier shall have a water vapour permeability not exceeding 1.13 ng/N-s (0.02 perm) and shall consist of one layer of aluminium foil and one layer of white paper laminate reinforced with fibreglass all factory laminated with flame retardant contact adhesive.
 - (iii) Plumbing Fixtures and Pipes
 - 1) All pipes chased in walls and floors need not be painted. All other piping, pipe hangers, supports, valves, equipment, steel work and all other works covered in this Specification whether in service ducts, within false ceilings or exposed, shall be painted as per the general specifications.
 - (iv) Cast Iron Pipework
 - 1) Where cast iron pipes run above false ceiling or buried below ground or subsoil, they shall be treated with two coats of bituminous paint to approval.

G1.5.10 Plant Identification

- (a) General
 - (i) This section includes identification, labelling and tagging requirements for piping, valves, instruments, equipment etc.
 - (ii) Colour coding for plant, equipment and pipework shall be as given in specifications.
- (b) Piping
 - (i) Labels shall be of the self-adhesive and self-sealed type and under all circumstances shall be adhered to pipework. Labels shall be easily replaceable without damaging the protective coating on the plant item. Labels shall withstand direct contact with all process chemicals, operating temperatures up to 120°C, and prolonged exposure to direct sunlight without deterioration for a minimum of five (5) years.
 - (ii) The Contractor at no extra cost to the Board shall replace defective or failed labels within the above parameters.

- (iii) Labels shall be provided at intervals not exceeding 5m and wherever necessary at all branches, at all junctions, changes in direction and both sides of valves, service appliances, fittings and wall penetrations. Where two or more pipes run in parallel, then the label for each pipe shall be at the same locations.
- (iv) For closed circuits the flow and return pipes shall be indicated by the use of the word "FLOW" on the one pipe and the word "RETURN" on the other.
- (v) Labels shall also be fitted to provide warnings or reminders of dangerous or potentially dangerous circumstances, and wherever called for elsewhere in the Specification.
- (vi) Information to be Included
 - 1) Unless otherwise specified the labels supplied shall indicate contents of pipeline, indicate direction of flow, origin and destinations of pipe and hazards due to pressure, temperature, chemical or physical properties of the contents as part of the legend.
- (vii) Label, Text Colours and Lettering Size
 - 1) Unless otherwise specified the base colour of the label shall be same as the colour of the pipework.
 - 2) The outside diameter of the pipe or insulation and jacketing determines the size of the letters and the maximum character field length. Unless otherwise specified, the label length, text height and colour of text respective to label colour are as follows:

Bare Pipe Size or OD. of Insulation	Text Height (mm)	Label Size (mm) Length of Colour Field
Smaller than 20 NB	6	75
20 to 32 NB	15	200
40 to 50 NB	20	200
65 to 150 NB	35	300
200 to 250 NB	65	600
Larger than 1250 NB	90	800

- 3) Label – Lettering Colours. Colour combinations of labels and lettering as follows:

Label	Text Colour
Green	White
Yellow	Black
Blue	White
Red	White

G1.5.11 Appendices

- (a) APPENDIX G1.5A - Specimen Schedule of Paint Systems and Protective Coatings.
- (b) APPENDIX G1.5B - Painting Systems for Concrete Surfaces.
- (c) APPENDIX G1.5C - Painting Systems for Building Works.
- (d) APPENDIX G1.5D - Painting Systems for Structural Steelwork.
- (e) APPENDIX G1.5E - Required Heat Resistant Coating Systems for Steelwork.
- (f) APPENDIX G1.5F - Painting Systems for Machinery, Equipment, Pipework and Associated Steelwork.
- (g) APPENDIX G1.5G - Colour Coding for Plant and Equipment (Potable Water and NEWater).
- (h) APPENDIX G1.5H - Identification Colours for All Pipes Including Underground Pipes.
- (i) APPENDIX G1.5I - Colour Coding for Plant and Equipment (Used Water).

END OF SECTION

APPENDIX G1.5A

SPECIMEN SCHEDULE OF PAINT SYSTEMS AND PROTECTIVE COATINGS

Surface to be coated	
General	<i>Component or section of the Works</i>
	<i>Environment category (BS EN ISO 12944 Parts 1 to 8) (BS EN ISO 14713 Parts 1 to 3) (BS 5493 – Table 1) (BS 6150 – Table 9)</i>
	<i>Type of coating system (BS EN ISO 12944 Parts 1 to 8) (BS EN ISO 14713 Parts 1 to 3) (BS 5493 – Table 2) (BS 6150 – Tables 5 to 8)</i>
	<i>Typical time to first maintenance (BS EN ISO 12944 Parts 1 to 8) (BS 6150 – Tables 10 to 15)</i>
Surface preparation prior to painting	<i>Type of surface treatment</i>
	<i>Quality level</i>
Metal coating	<i>Type of metal coating</i>
	<i>Thickness of coating</i>
Surface preparation prior to metal coating	<i>Type of surface treatment</i>
	<i>Quality level</i>
Type of paint	<i>Priming coat</i>
	<i>Stripe coat</i>
	<i>Undercoat</i>
	<i>Finish</i>
Number of coats and DFT	<i>Priming coat</i>
	<i>Stripe coat</i>
	<i>Undercoat</i>
	<i>Finish</i>
Colour	<i>Priming coat</i>
	<i>Stripe coat</i>
	<i>Undercoat</i>
	<i>Finish</i>

APPENDIX G1.5B

PAINTING SYSTEMS FOR CONCRETE SURFACES

Surface	Location	Preparation	Primer	DFT	Undercoat	DFT	Finish Coat	DFT
Concrete faces external to buildings and structures SYSTEM 1	Not submerged and not exposed to aggressive environments.	Removal of all efflorescence, cleaning, anti-fungal wash and degreasing	Styrene acrylic based sealer	25	Flexible acrylic water based non plasticised elastomeric	350	Flexible acrylic water based	50
Concrete faces within buildings and structures SYSTEM 2	In non-aggressive environments	Removal of all efflorescence, cleaning, anti-fungal wash and degreasing	PVA acrylic copolymer matt emulsion. Minimum Solids Contents 45 to 50 %	50	PVA acrylic copolymer matt emulsion - Minimum Solids Contents 45 to 50 %	50	PVA acrylic copolymer matt emulsion - Minimum Solids Contents 45 to 50%	50
Concrete tanks, channels, Internal surfaces of concrete drains SYSTEM 3	In chemically corrosive environment	Removal of all efflorescence, cleaning, anti-fungal wash and degreasing	Epoxy amine cured sealer Minimum Solids Contents 50%	50	Bitumen amine cured epoxy, two pack (brown) Minimum Solids Contents 75%	350	Bitumen amine cured epoxy, two pack (black) Minimum Solids Contents 75%	350
Concrete faces within buildings and structures SYSTEM 4	In chemically corrosive environments including bunds, channels at chemical dosing points	Removal of all efflorescence, grease etc. Grit blasting or scarification	Epoxy amine cured 50 sealer Minimum Solids Contents 50%	50	-	-	High build, two 500 pack Epoxy Novalac system such as Fosroc Nitocote EN901 or similar approved applied by roller or airless spray. Multiple coats required to achieve minimum DFT of 500 microns	500

Surface	Location	Preparation	Primer	DFT	Undercoat	DFT	Finish Coat	DFT
Concrete faces within buildings and structures SYSTEM 5	In chemically corrosive locations	Removal of all efflorescence, grit blasting or scarification	Vinyl ester/peroxide penetrating primer	50	Trowel applied silica filled vinyl ester base coat with glass fibre mat 0.5 kg/m ² saturated with vinyl ester	1500	Glass flake reinforced vinyl ester	500
Note: DFT units in micron.								

APPENDIX G1.5C

PAINTING SYSTEMS FOR BUILDING WORKS

Surface	Location	Finish	Preparation	Primer	DFT	Undercoat	DFT	Finish Coat	DFT
Hardwood SYSTEM 6	Interior and Exterior	Gloss	Filling, stopping, and knotting; sanding down to smooth finish	Aluminium priming paint to BS 7956	30	Alkyd based lead free high opacity undercoat	40	Oil modified alkyd resin enamel Coat 1 Ditto Coat 2	30 30
Softwood, primed or unprimed SYSTEM 7	Interior and exterior	Gloss	Filling, stopping, and knotting; sanding down to smooth finish	Aluminium priming paint to BS 7956	30	Alkyd based lead free high opacity undercoat	40	Oil modified alkyd resin enamel Coat 1 Ditto Coat	30 30
Hard SYSTEM 8	Exterior	Gloss	Zinc chromate spot primer to screw and nail heads	Stabilising primer	30	Alkyd based lead free high opacity undercoat	40	Oil modified alkyd resin enamel Coat 1 Ditto Coat 2	30 30
Hardboard SYSTEM 9	Interior	Flat	Zinc chromate spot primer to screw and nail heads	PVA acrylic copolymer matt emulsion	50	PVA acrylic copolymer matt emulsion	50	PVA High opacity acrylic emulsion	50
Plaster including plaster board SYSTEM 10	Interior	Flat	Removal of all efflorescence, filling with plaster based filler; degreasing and sanding down	PVA acrylic copolymer matt emulsion	50	PVA acrylic copolymer matt emulsion	50	PVA acrylic copolymer matt emulsion	50
Mortar rendering to concrete, blockwork or brickwork SYSTEM 11	Interior and Exterior	Flat	Removal of all efflorescence, cleaning and degreasing	PVA acrylic copolymer matt emulsion primer	50	PVA acrylic copolymer matt emulsion	50	PVA acrylic copolymer matt emulsion	50
Mortar rendering to concrete, blockwork or brickwork SYSTEM 12	Exterior	Textured	Stiff brush cleaning	Stabilising primer	40	Textured emulsion masonry paint	70	Textured emulsion masonry paint	70

Surface	Location	Finish	Preparation	Primer	DFT	Undercoat	DFT	Finish Coat	DFT
Galvanised miscellaneous metal. See Note. SYSTEM 13	Interior and exterior	Gloss	Etch prime	To BS 3698	30	Alkyd based lead free high opacity undercoat	40	Oil modified alkyd resin enamel Coat 1 Ditto Coat 2	30 30
Note: 1. Unless specified otherwise or directed by the S.O., all steel and ferrous metals incorporated in builder's work shall be galvanised prior to being painted. DFT units in micron.									

APPENDIX G1.5D

PAINTING SYSTEMS FOR STRUCTURAL STEELWORK

Surface	Location	Finish	Preparation	Primer	DFT	Undercoat	DFT	Finish Coat	DFT
Structural Steel (Factory Galvanised) SYSTEM 14	Above Water Level	Gloss	Etch Prime or T Wash	Zinc Phosphate /Chromate Alkyd Primer	50	High Build Chlorinated Rubber	75	High Build Chlorinated rubber	75
Structural Steel (Factory Zinc Sprayed) SYSTEM 15	Above Water Level	Gloss	Etch Prime	Zinc Phosphate /Chromate Alkyd Primer	50	High Build Chlorinated rubber	75	High Build Chlorinated Rubber	75
Structural Steel (Factory blast cleaned and blast primed) SYSTEM 16	Above Water Level	Gloss	Blast Clean to Sa2½ followed immediately by 25 microns DFT epoxy iron oxide two pack blast primer applied in the factory	Epoxy Zinc Chromate / Phosphate Primer, two pack	75	High Build Cured Epoxy coating, two pack	200	Two pack aliphatic polyurethane finish	50
Structural Steel (Factory blast cleaned and primed) SYSTEM 17	Cast into or encased in concrete	-	Blast Clean to Sa2½ in factory conditions followed immediately by application of primer	Inorganic Zinc Silicate Primer applied in factory conditions	50				
Notes: 1. Unless specified otherwise or directed by the S.O., all miscellaneous steel and ferrous metals (excluding structural steel) cast into or encased in builder's work shall be galvanised prior to being painted. 2. ABOVE WATER LEVEL applies to Steelwork not in contact with used water, sludge or water to be used for drinking, washing or cooking 3. BELOW WATER LEVEL applies to Steelwork in contact with water to be used for drinking, washing or cooking. 4. DFT units in micron.									

APPENDIX G1.5E

REQUIRED HEAT RESISTANT COATING SYSTEMS FOR STEELWORK

WORKING TEMPERATURE	SURFACE TREATMENT	FIRST COAT	SECOND COAT	THIRD COAT
50°C to 175°C	Blast clean to BS 4232 2nd quality	Polyvinyle Butyral (25 micron) Zinc Chromate Primer (15 – 25 micron)	Aluminium heat resistant @ 200°C (25 micron)	Aluminium heat resistant @ 200°C (25micron)
175° to 500°C	Blast clean to BS 4232 2nd quality	Aluminium pigmented silicone heat resistant (25micron)	Aluminium pigmented silicone heat resistant (25 micron)	
	NB: A minimum temperature of 350°C within a short time after application is required.			

APPENDIX G1.5F

PAINTING SYSTEMS FOR MACHINERY, EQUIPMENT, PIPEWORK AND ASSOCIATED STEELWORK

Surface	Location	Finish	Preparation	Primer	DFT	Undercoat	DFT	Finish Coat	DFT
Steel, Cast Iron, and Ductile Iron SYSTEM 18	Above Water Level	Gloss	Blast Clean to Sa2½ followed immediately by 50 microns DFT zinc phosphate / chromate alkyd blast primer			Zinc Phosphate Alkyd Primer	50	High Build Chlorinated rubber	75
Steel (Factory Galvanised) SYSTEM 19	Below Water Level		Etch Prime or T Wash	High Build Epoxy Primer	125	High Build Epoxy	50	High Build Epoxy	75
Steel, Cast Iron, and Ductile Iron, Zinc Sprayed SYSTEM 20	Below Water Level		T Wash	High Build Epoxy Primer	125	High Build Epoxy	50	High Build Epoxy	75
Steel Doors SYSTEM 21	Roller Doors in Chemically Aggressive Environments		As Recommended by the manufacturer to suit Environmental Exposure Conditions						
Pipework SYSTEM 22	General	Gloss	Blast Clean to Sa2½ followed immediately by 25 microns DFT epoxy iron oxide two pack blast primer	Two pack polyamide cured Epoxy Zinc Chromate / Phosphate Primer	75	Two pack high build amine adduct cured epoxy	100	Two pack aliphatic polyurethane	50
Valves and Penstocks SYSTEM 23	Below Water Level		Blast Clean to Sa2½ followed immediately by 25 microns DFT epoxy iron oxide two pack blast primer	Epoxy Zinc Chromate / Phosphate Primer, Two Pack	75	Coal Tar Cured Epoxy Coating, Two Pack	100	Coal Tar Cured Epoxy Coating, Two Pack	150
Special Applications SYSTEM 24	Exterior	Gloss	Blast Clean to Sa2½ followed immediately by 50 microns DFT Inorganic Zinc Silicate Primer			Epoxy sealer coat (red)	75	Glass flake epoxy (grey)	350

Surface	Location	Finish	Preparation	Primer	DFT	Undercoat	DFT	Finish Coat	DFT
Pipework Special Coating SYSTEM 25	Interior lining and exterior coating				100% solids, thermo setting fusion bonded , dry powder epoxy	750			
Pipes and Valves, factory bituminous coated SYSTEM 26	In Chambers and Tanks	Gloss	Stiff wire brushing	Aluminium seal coat	30	Alkyd chlorinated rubber	40	Alkyd chlorinated rubber	40
Switchboards, Shells, Frames and Backplates SYSTEM 27	Steel	Gloss	As Specified						
Notes: 1. Unless specified otherwise or directed by the S.O., all miscellaneous steel and ferrous metals (excluding structural steel) cast into or encased in builder's work shall be galvanised prior to being painted. 2. ABOVE WATER LEVEL applies to Machinery and Steelwork not in contact with sewage, sludge or water to be used for drinking, washing or cooking. 3. BELOW WATER LEVEL applies to Machinery and Steelwork in contact with water to be used for drinking, washing or cooking. 4. DFT units in micron									

APPENDIX G1.5G

COLOUR CODING FOR PLANT AND EQUIPMENT (POTABLE WATER & NEWATER)

EQUIPMENT	COLOUR FINISH	BS 4800 SHADE No
Steelwork and Equipment 500 mm above water level		
Raw Water pumps, motor stools, pipework and valves	Jungle Green	12-D-45
Treated water pumps, pipes, valves and motor stools	Dark Blue	18-E-53
Steelwork and Equipment not in contact with sewage, sludge or potable water		
Dry Well Motors	Tangerine	06-E-51
Guards	Flame	04-E-53
Gearboxes	Saxe Blue	18-C-39
Compressors & Blowers	Blue	20-E-51
Lifting Gentries, Cranes	Yellow	08-E-53
Crane Hooks runway beams	Blue	18-E-53
Steelwork and Equipment Below Water Level		
Machinery & Steelwork to include Pumps, Penstocks, Pipework etc.	Black	00-E-53
Electrical Equipment		
Engines and Generators	Blue	18-E-51
Air Compressors	Blue	18-E-45
Switchboard Shells	Sand Beige	5Y7/1
Switchboards Supporting Frames	Black	00-E-53
Switchboard internal equipment mounting plates	White	00-E-55
Control and instrument panels - external	Sand Beige	5Y7/1
Control and instrument panels - internal	White	00-E-55
Motor Control Centres and Switchboards	Green Satin	14-E-51
Cable Trays	Yellow	10-E-53
Control Conduits	Suede	04-B-25
Electrical services i.e. Conduits and Trunkings	Tangerine	06-E-51
GRP Covers (Pigmented)	Spruce	14-C-39
Ventilation and Air Conditioning Equipment		
Chillers	Fiesta Blue	18 D-43
Chilled Water Pumps	Island Green	12D43
Packaged Units	Pewter	00-A-05
Chilled Water Piping	Island Green	BS 12D43
Chilled Water Expansion Tank	Island Green	BS 12D43
Equipment Drains	Dark Green	14-E-53
Vent Piping	Ice Blue	BS 7-081

EQUIPMENT	COLOUR FINISH	BS 4800 SHADE No
Warning Sign	Signal Red	BS 04E.53
Receiver for Refrigerant	Salmon	8-E-51
Ventilation/Exhaust Duct	Magnolia	8-B-17
Ventilation/Exhaust Fans	Magnolia	8-B-17
Valves coded the same colour as the pipe line		
Storage Tanks/ Vessels		
Diesel Storage Tanks	Grey	18-B-19
Lube Oil Storage Tanks	Brown	06-C-39
Air Receivers	Blue	18-E-51
Acid Tanks/ Pumps	Orange	06-E-55
Alkali Storage Tanks/ Pumps	Violet	22-C-37
Hypochlorite Tanks/ Pumps	Violet	22-C-37
Bisulphite Tanks/ Pumps	Violet	22-C-37

TUAS WATER RECLAMATION PLANT
CONTRACT 4B – DIGESTION PRE-TREATMENT AND
SIDESTREAM NITROGEN REMOVAL FACILITIES

VOLUME 2

APPENDIX G1.5H

IDENTIFICATION COLOURS FOR ALL PIPES INCLUDING UNDERGROUND PIPES

PIPE CONTENTS	CCCC	BASE COLOUR	BS 4800 / BS 5252 SHADE NO.
Aeration Air	AA	Light Blue	18 E 51
Air High Pressure	AHP	Light Blue	20 E 51
Air Low Pressure	ALP	Blue	18 E 49
Ammonium Hydroxide	NHH	Blue	18 E 49
Ammonium Sulphate	NHS	Lettuce Green	12 E 55
Antiscalant	ASC	Jade	16 D 45
Bioscrubber Liquor	BL	Dark Green	14 D 45
Carbon Dioxide (Gas)	CDG	Grey	00 A 07
Chilled Water Return	CHWR	Island Green	12 D 43
Chilled Water Supply	CHWS	Island Green	12 D 43
CIP Return	CIPR	Light Blue	20 E 50
CIP Supply	CIPS	Light Blue	20 E 50
Citric Acid	CIT	Orange	06 E 55
Condensate	SC	Dark Green	14 D 45
Dewatering Centrate Recycle	DC	Dark Green	14 D 45
Digested Sludge	DS	Light Blue	18 E 41
Domestic Used Water	DUW	Green	14 E 53
Ferric Chloride	FC	Light Green	14 C 31
Fire Water	FPW	Red	04 E 55
Food Waste	FW	Black	00 E 53
Foul (Sanitary) Drain	FD	Black	00 E 53
Foul Air	FA	White	00 E 55
Fuel Oil (Return)	FOR	Dark Purple	24 C 37
Fuel Oil (Supply)	FOS	Dark Purple	24 C 37
Gas (Biogas)	GB	Canary Yellow with Red Banding	10 E 55
Gas (Mains)	GM	Canary Yellow	10 E 55
Greasy Waste	GW	Black	00 E 53
Grit Slurry	GR	Black	00 E 53
High Pressure Hot Water	HPHW	Cobalt Blue with Red Banding	18 E 53
Hot Water (Return)	HWR	Yellow Red	06 E 51
Hot Water (Supply)	HWS	Yellow Red	06 E 51
Industrial Used Water	IUW	Green	14 E 53
Industrial Water	IW	Yellow	08 E 51
Instrument Air	IA	Light Blue	18 E 51
Lime Hydrated	LH	Light Green	14 E 50
Lime Sludge	LSL	Light Green	14 E 50
Lime Solution	LS	Light Green	14 E 50

PIPE CONTENTS	CCCC	BASE COLOUR	BS 4800 / BS 5252 SHADE NO.
Low Pressure Hot Water	LPHW	Yellow Red	06 E 45
Membrane Filtrate	MF	Light Blue	20 E 50
NEWater	NW	Purple	22 C 37
Plant Drain	PD	Dark Green	14 D 45
Plant Effluent	PLE	Cobalt Blue	18 E 53
Plant Water (DLM MF)	G2D	Cobalt Blue	18 E 53
Plant Water (ILM MF)	G2I	Cobalt Blue with black banding	18 E 53
Plant Water High Pressure	G2HP	Cobalt Blue	18 E 53
Polyelectrolyte	PO	Grey	00 A 07
Primary Effluent	PE	Green	14 E 53
Primary Influent	PI	Green	14 E 53
Primary Scum	PSM	Yellow Red	08 D 41
Primary Sludge Domestic	PSD	Black	00 E 53
Primary Sludge Industrial	PSI	Black	00 E 53
Return Activated Sludge	RAS	Brown	08 D 45
RO Concentrate	ROC	Cobalt Blue	18 E 53
RO Feed	ROF	Light Blue	20 E 50
RO Permeate	ROP	Light Green	14 E 50
Sample	S	White	00 E 55
Scour Air	AS	Light Blue	18 D 51
Screenings	SG	Yellow Red	08 D 41
Secondary Scum	SSM	Yellow Red	08 D 41
Sodium Bisulphite	SBS	Violet	22 B 27
Sodium Hydroxide	SHYD	Purple	24 E 53
Sodium Hypochlorite	SHYP	Purple	24 E 53
Steam High Pressure	STHP	Dawn Grey	10 A 03
Steam Low Pressure	STLP	Dawn Grey	10 A 03
Sulphuric Acid	SA	Purple Blue	20 C 33
Thickened Primary Sludge	TPS	Black	00 E 53
Thickened Waste Activated Sludge	TWAS	Brown	08 D 45
Thickening Centrate Domestic	TCD	Dark Green	14 D 45
Thickening Centrate Industrial	TCI	Dark Green	14 D 45
Waste Activated Sludge Domestic	WASD	Brown	08 D 45
Waste Activated Sludge Industrial	WASI	Brown	08 D 45

Note:

The above paint colour shall be subject to final confirmation by the S.O.. The Contractor shall submit the colour shades for the S.O.'s approval 2 weeks in advance of any painting works to be carried out on piping.

APPENDIX G1.5I

COLOUR CODING FOR PLANT AND EQUIPMENT (USED WATER)

Description	Basic Colour	BS 4800 / BS 5252 Ref. No.	Finish	Remarks
Actuators (hydraulic & electric)	Yellow	10.C.35	Gloss	-
Automatic Drip Traps	Red	04.E.56	Gloss	
Air Separator	Light Blue	18.B.17	Gloss	
Air Handling Units	Yellow	10.A.03	Gloss	
Activated Carbon Units	Yellow Red	08.C.31	Gloss	
Aeration Blower/Motor/Enclosure	Blue	18.E.49	Gloss	
Automatic Water Strainers	Blue	18.E.53	Gloss	
Air Receivers	Purple Blue	20.E.51	Gloss	
Bioreactor Transfer Pump	Yellow Red	08.D.41	Gloss	
Bioreactor Anoxic Mixers (above TWL)	Blue	18.E.50	Gloss	
Boiler Heat Exchanger	Blue Grey	18.B.23	Gloss	
Centrifuge Collection Screw Conveyor	No Painting			Stainless Steel
Cake Bin Feed Screw	No Painting			Stainless Steel
Chilled Glycol and Dehumidification System	Light Blue	18.E.50	Gloss	
Chemical Feeder	Light Blue	18.B.19	Gloss	
Coarse Bubble Diffusion System	No Painting	-	-	CPVC
Chilled Water Expansion Tanks	Green Yellow	12.D.43	Gloss	
Cake Distribution Screw Conveyor	No Painting			Stainless Steel
Dewatering Centrifuge	Grey Green	12.B.21	Gloss	-
Dewatering Centrifuge in Line Mixer	Grey Green	12.B.21	Gloss	-
Dewatering Centrifuge Main Drive	Grey Green	12.B.21	Gloss	-
Dewatering Centrifuge Back Drive	Grey Green	12.B.21	Gloss	-

Description	Basic Colour	BS 4800 / BS 5252 Ref. No.	Finish	Remarks
Dewatering Centrifuge Lube Oil System	Grey Green	12.B.21	Gloss	-
Dry Polymer Bulk Unloader	No Painting			Factory
Dry Polymer Storage Silo	No Painting			Stainless Steel
Dry Polymer Baghouse	No Painting			Factory
Dry Polymer Volumetric Screw Feeder	No Painting			Stainless Steel
Dry Polymer Recharge Hopper	No Painting			Stainless Steel
Dry Polymer Loss in Weight Feeder	No Painting			Stainless Steel
Dry Polymer Eductor Hopper	No Painting			Stainless Steel
Dry Polymer Eductor	No Painting			Stainless Steel
Dry Polymer Vacuum Exhauster	Red	04.D.43	Gloss	-
Dry Polymer Vacuum Rotary Valve	Red	04.D.43	Gloss	-
Dry Polymer Vacuum Separator/Baghouse	Red	04.D.43	Gloss	-
Dewatered Cake Storage Hopper	No Painting			Stainless Steel
Dewatered Cake Bin	No Painting			Stainless Steel
Dumpster	Blue	18. E.53	Gloss	
Dust Dense Phase Pneumatic Conveyance System	Green	14.E.56	Gloss	
Dry Vacuum System Exhauster	Brown	08.D.44	Gloss	
Dry Vacuum System Separator/Baghouse	Brown	08.D.44	Gloss	
Dry Vacuum System Rotary Valve	Brown	08.D.44	Gloss	
Dry Vacuum System Bulk Bag Loader	Brown	08.D.44	Gloss	
Digester Transfer Pumps	Light Brown	08.D.44	Gloss	
Digester Scum Transfer Pumps	Yellow Red	08.D.43	Gloss	
Digester Mixing Pumps	Light Brown	08.D.44	Gloss	

TUAS WATER RECLAMATION PLANT
CONTRACT 4B – DIGESTION PRE-TREATMENT AND
SIDESTREAM NITROGEN REMOVAL FACILITIES

VOLUME 2

Description	Basic Colour	BS 4800 / BS 5252 Ref. No.	Finish	Remarks
Digester Heating Sludge Pumps	Light Brown	08.D.44	Gloss	
Digester Gas Burners	No Painting			Stainless Steel
Digester Gas Boosters	Red	04.E.56	Gloss	
Emergency Flood Pump	Black	00.E.53	Gloss	
Expansion Tank	Blue Grey	18.B.23	Gloss	
Effluent Pumps/Booster Pump	Blue	18.E.53	Gloss	
Furnace and Burner System	Purple Blue	20.E.56	Gloss	
Final Product Loadout Chute	No Painting			Stainless Steel
Fuel Oil Transfer Pumps	Purple	24.C.37	Gloss	
Fine Bubble Diffusion System	No Painting	-	-	CPVC
Ferric Chloride Storage Tanks	Light Green	14.C.31	Gloss	
Ferric Chloride Dosing Pumps	Light Green	14.C.31	Gloss	
Gas Moisture Sediment Traps	No Painting			Stainless Steel
Gas Storage Tank Floating Covers	No Painting			Stainless Steel
Gas Purifiers	No Painting			Stainless Steel
Grade 2 Plant Water Booster Pumps	Light Blue	20.E.50	Gloss	
Grit Removal Basin Equipment	Black	00.E.53	Gloss	
Grit Pump	Black	00.E.53	Gloss	
Grit Separator	No Painting	-		Stainless Steel
Grit Classifier	No Painting	-	-	Stainless Steel
Grit Conveyors	No Painting	-	-	Stainless Steel
Grit Storage Silos	No Painting	-	-	Stainless Steel
Grade 1 Plant Water Pumps	Light Green	14.E.50	Gloss	

Description	Basic Colour	BS 4800 / BS 5252 Ref. No.	Finish	Remarks
Grade 2 Plant Water Pumps	Light Blue	20.E.50	Gloss	
Grade 3 Plant Water Pumps	Blue	18.E.49	Gloss	
G1 Water Storage Tank	Light Green	14.E.50	Gloss	
G2 Water Storage Tank	Light Blue	20.E.50	Gloss	
Hydraulic Power Packs	Blue	20.E.51	Gloss	
Hot Water Supply Pumps	Blue Grey	18.B.23	Gloss	
Hot Water Boilers	Blue Grey	18.B.23	Gloss	
Hot Water Pumps	Blue Grey	18.B.23	Gloss	
Isolation Valves	Yellow	10.E.55	Gloss	-
Influent Pump	Green	14.E.53	Gloss	
Influent Pump Motors	Blue	18.E.51	Gloss	
Induced Draft Fan	Neutral	00.A.03	Gloss	
Instrumentation/Control Panel	Grey	10.A.05	Gloss	
Influent Fine Screen (above TWL)	No Painting	-	-	Stainless Steel
Jib Crane, Bridge Crane and Monorail Girders, Trolleys, & Hoists	Yellow Red	08.E.51	Gloss	-
Jib Crane, Bridge Crane and Monorail Hooks	Red	04.E.53	Gloss	-
Bridge Crane Runway Beams	Blue	18.B.25	Gloss	-
Large Knife Gate Valves	No Painting	-	-	Stainless Steel
Lift Sump Pump	Black	00.E.53	Gloss	
Main Transport Screw Conveyor 1	No Painting			Stainless Steel
Main Transport Screw Conveyor 2	No Painting			Stainless Steel
Mixer	No Painting			Stainless Steel
Mineral Oil Storage Tank	Light Blue	18.B.19	Gloss	

TUAS WATER RECLAMATION PLANT
CONTRACT 4B – DIGESTION PRE-TREATMENT AND
SIDESTREAM NITROGEN REMOVAL FACILITIES

VOLUME 2

Description	Basic Colour	BS 4800 / BS 5252 Ref. No.	Finish	Remarks
Mineral Oil Feed System	Light Blue	18.B.19	Gloss	
Mineral Oil Unloading Pump	Light Blue	18.B.19	Gloss	
Module Water Pumps	Blue	18.E.53	Gloss	
Module Drain Pumps	Dark Green	14.D.45	Gloss	
Microfiltration Feed Pump	Blue	18.E.53	Gloss	
Make-up Water Tanks	Green Yellow	12.C.33	Gloss	
Odour Extraction Fans	Yellow Red	08.C.31	Gloss	
Oil / Grease Removal Equipment (above TWL)	Black	00.E.53	Gloss	
Polymer Mix/feed tanks	No Painting			Stainless Steel
Polymer Mixing Tank Wetting Head	No Painting			Stainless Steel
Pumps & Motors (general)	Same colour as pipework of fluid conveyed unless indicated otherwise	See Section 15060, Process Piping - General, Pipe Schedule	Gloss	-
Preseparator/Polycyclone	Green Yellow	12.E.55	Gloss	
Product Cooler	Blue Green	16.E.53	Gloss	
Product Dense Phase Pneumatic Conveyance System	Tan	10.B.21	Gloss	
Product Storage Silo	No Painting			Stainless Steel
Product Loadout Screw Conveyor	No Painting			Stainless Steel
Process Baghouse	No Painting			Stainless Steel
Process Baghouse Rotary Valve	No Painting			Stainless Steel
Process Baghouse Exhauster	Light Yellow	10.D.41	Gloss	
Propane Tanks	Red	04.E.53	Gloss	
Primary Clarifier Equipment (above TWL)	Black	00.E.53	Gloss	

Description	Basic Colour	BS 4800 / BS 5252 Ref. No.	Finish	Remarks
Primary Grit and Sludge Pump	Black	00.E.53	Gloss	
Primary Grit Separator	No Painting	-	-	Stainless Steel
Primary Grit Classifier	No Painting	-	-	Stainless Steel
Module Water Pumps	Blue	18.E.53	Gloss	
Module Drain Pumps	Dark Green	14.D.45	Gloss	
Microfiltration Feed Pump	Blue	18.E.53	Gloss	
Make-up Water Tanks	Green Yellow	12.C.33	Gloss	
Odour Extraction Fans	Yellow Red	08.C.31	Gloss	
Oil / Grease Removal Equipment (above TWL)	Black	00.E.53	Gloss	
Polymer Mix/feed tanks	No Painting			Stainless Steel
Polymer Mixing Tank Wetting Head	No Painting			Stainless Steel
Pumps & Motors (general)	Same colour as pipework of fluid conveyed unless indicated otherwise	See Section 15060, Process Piping - General, Pipe Schedule	Gloss	-
Preseparator/Polycyclone	Green Yellow	12.E.55	Gloss	
Product Cooler	Blue Green	16.E.53	Gloss	
Product Dense Phase Pneumatic Conveyance System	Tan	10.B.21	Gloss	
Product Storage Silo	No Painting			Stainless Steel
Product Loadout Screw Conveyor	No Painting			Stainless Steel
Process Baghouse	No Painting			Stainless Steel
Process Baghouse Rotary Valve	No Painting			Stainless Steel
Process Baghouse Exhauster	Light Yellow	10.D.41	Gloss	
Propane Tanks	Red	04.E.53	Gloss	

TUAS WATER RECLAMATION PLANT
CONTRACT 4B – DIGESTION PRE-TREATMENT AND
SIDESTREAM NITROGEN REMOVAL FACILITIES

VOLUME 2

Description	Basic Colour	BS 4800 / BS 5252 Ref. No.	Finish	Remarks
Primary Clarifier Equipment (above TWL)	Black	00.E.53	Gloss	
Primary Grit and Sludge Pump	Black	00.E.53	Gloss	
Primary Grit Separator	No Painting	-	-	Stainless Steel
Primary Grit Classifier	No Painting	-	-	Stainless Steel
Primary Grit Conveyor	No Painting	-	-	Stainless Steel
Primary Grit Silo	No Painting	-	-	Stainless Steel
Primary Scum Pumps	Yellow Red	08.D.41	Gloss	
Packaged Units	Neutral	00.A.05	Gloss	
Recycle Bin (w/ bin activator)	No Painting			Stainless Steel
Recycle Dosing Screw	No Painting			Stainless Steel
Rotary Dryer	No Painting			Stainless Steel
Regenerative Thermal Oxidiser	Purple	24.E.53	Gloss	
Rotary Valve (for preseparator and at screener inlet)	Red Purple	02.E.56	Gloss	
Recycle/Product Screen	Green Yellow	12.D.43	Gloss	
Roller Mill	Orange Red	06 E 55	Gloss	
Recycle Cooler	No Painting			Stainless Steel
Recycle Transport Screw	No Painting			Stainless Steel
Recycle Screw Conveyor	No Painting			Stainless Steel
Recycle Bucket Elevator	No Painting			Stainless Steel
Rotary Valve	No Painting			Stainless Steel
Recycle Feed Screw	No Painting			Stainless Steel
Return Screw Conveyor	No Painting			Stainless Steel
RAS Pumps	Brown	08.D.45	Gloss	

Description	Basic Colour	BS 4800 / BS 5252 Ref. No.	Finish	Remarks
RO Feed Pumps	Blue	18.E.53	Gloss	
RO Modules	Blue	18.E.53	Gloss	
Saturator (plate-type w/ venturi scrubber)	No Painting			Stainless Steel
Saturator Exhaust Fan	Gray	00.A.09	Gloss	
Surge Bin	Yellow Red	08.E.55	Gloss	
Silo Baghouse	No Painting			Stainless Steel
Silo Baghouse Rotary Valve	No Painting			Stainless Steel
Silo Baghouse Exhauster	Orange	08.E.55	Gloss	
Sump Pumps	Black	00.E.53	Gloss	
Sludge Heat Exchangers	Light Brown	08.D.44	Gloss	
Screenings Conveyor	No Painting	-	-	Stainless Steel
Screenings Press	No Painting	-	-	Stainless Steel
Screenings Hopper	No Painting	-	-	Stainless Steel
Scum Concentrator	Yellow Red	08.D.41	Gloss	
Submersible Scum Mixer	Black	00.E.53	Gloss	
Scum Transfer Pumps	Yellow Red	08.D.41	Gloss	
Secondary Clarifier Equipment (above TWL)	Black	00.E.53	Gloss	
Sodium Hypo Chlorite Storage Tank	Purple	20.E.53	Gloss	
Sodium Hypo Chlorite Dosing Pumps	Purple	20.E.53	Gloss	
Sulphuric Acid Storage Tanks	Purple Blue	20.C.33	Gloss	
Sulphuric Acid Transfer / Dosing Pumps	Purple Blue	20.C.33	Gloss	
Sodium Hydroxide Storage Tanks	Violet	22.C.33	Gloss	
Sodium Hydroxide Transfer / Dosing Pumps	Violet	22.C.33	Gloss	

TUAS WATER RECLAMATION PLANT
CONTRACT 4B – DIGESTION PRE-TREATMENT AND
SIDESTREAM NITROGEN REMOVAL FACILITIES

VOLUME 2

Description	Basic Colour	BS 4800 / BS 5252 Ref. No.	Finish	Remarks
Sprinkler Storage Tank	Red	04.E.55	Gloss	
Transmission shafts, rotating parts and guards for rotating parts	Red	04.E.53	Gloss	-
Thickening Centrifuge	Dark Brown	08.B.29	Gloss	-
Thickening Centrifuge in Line Mixer	Dark Brown	08.B.29	Gloss	-
Thickening Centrifuge Main Drive	Dark Brown	08.B.29	Gloss	-
Thickening Centrifuge Back Drive	Dark Brown	08.B.29	Gloss	-
Thickening Centrifuge Lube Oil System	Dark Brown	08.B.29	Gloss	
Thickened Sludge Conveyor	No Painting			Stainless Steel
Thickened Sludge Booster Pumps	Light Brown	08.D.44	Gloss	
Upper Product Loadout Chute	No Painting			Stainless Steel
Valves	Same colour as pipework of fluid conveyed unless indicated otherwise		Gloss	-
Ventilation Fans	Neutral	00.A.03	Gloss	
Wet Material Dosing Screw	No Painting			Stainless Steel
Wet Material Transfer Screw	No Painting			Stainless Steel
WAS Pumps	Brown	08.D.45	Gloss	
Wet Chemical Scrubbers	Yellow Red	08.C.31	Gloss	
22kV HT Panels	Orange	06.E.53	Gloss	-
6.6kV HT Panels	Yellow	10.E.53	Gloss	-
400V/230V LT Panels	Grey	00.A.05	Gloss	-